Rhythmic Variance Control at Sleep Spindles: A Three-Dataset Replication of Timing-Locked State-Space Reconfiguration Without Embodied-Anchor Moderation

Spindle-locked geometry change replicates across ANPHY, BOAS, and RichSleep; a dual-arm cardiorespiratory anchor test finds no moderation

Article type: Research Article

Short title: RVC at sleep spindles: three-dataset replication

Keywords: sleep spindles, state-space geometry, rhythmic variance control, embodied anchoring, cardiorespiratory coupling, replication

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Abstract Sleep spindles are proposed to act as rhythm-defined "variance-control" events that transiently reconfigure cortical state-space geometry. We tested this using Noetic Diffusion Theory's stratified 9D state-space projection (Meta-Noetic Phase Space, MNPS) in three independent overnight PSG datasets (ANPHY, $N = 27$; BOAS/ds005555, $N = 128$; RichSleep, $N = 24$; total 179 subjects), aligning spindle-locked windows against matched non-spindle N2 controls and a within-spindle pre-event baseline. A subject-weighted timing contrast (event vs. pre-spindle baseline) showed a reproducible increase in the deviation-from-integration-segregation-balance endpoint d_n in all three datasets (ANPHY $+0.106$, $q = 0.009$; BOAS $+0.165$, $q = 0.011$; RichSleep $+0.29$, $q = 0.0006$), with the entropy-shape endpoint e_s supportive in two of three. A broader event-vs-matched-control contrast was also significant on most endpoints but proved baseline-sensitive under decomposition, so the timing contrast is treated as the primary claim. We then tested whether pre-event embodied anchor (EAP) state moderates this effect. In ANPHY (cardiac-only anchor) and BOAS (respiratory-only anchor, missing in 66/128 subjects), 53/54 moderation tests were null. RichSleep's simultaneous, 100%-covered cardiac+respiratory composite anchor removed the single-channel-missingness objection and reproduced the same null across all 9 endpoints in both delta modes. Anchor-augmented sensitivity geometry showed the expected construction-validity pattern (only the directly-modified endpoint moved) with zero flips in FDR status. Two additional, more mechanistically direct RichSleep tests - spindle-onset cardiac/respiratory phase-locking (0/48 within-subject and 0/2 population-level Rayleigh tests significant), and heartbeat-evoked-potential-amplitude moderation (0/18 tests FDR-significant) - were also null. We conclude that spindle-locked d_n reconfiguration is a replicable, dataset-independent timing effect, and that it is not moderated by pre-event cardiorespiratory anchor state under any anchor or mechanism tested here.

Author Summary

Brief bursts of brain activity called sleep spindles are thought to briefly reorganize how the brain's activity is structured. We measured this reorganization with a geometric state-space method across three independent overnight sleep recordings (179 people total) and found the same effect each time: one particular geometric signature rises right at the spindle and falls back afterward. We also asked whether the body's heart-rate and breathing state just before a spindle changes the size of this effect, using the richest of the three datasets - which uniquely combines simultaneous heart and breathing signals - and found no evidence that it does.

1. Introduction

Sleep spindles - 0.5-2 s bursts of 11-16 Hz thalamocortical oscillation during N2 sleep - are one of the best-characterized rhythmic events in human electrophysiology and are widely linked to memory consolidation and thalamocortical gating (Vallat & Walker, 2021). Within Noetic Diffusion Theory (NDT), spindles have been proposed as candidate **Rhythmic Variance Control** (RVC) events: rhythm-defined windows in which a transient increase in gating precision should show up as a measurable, reproducible reconfiguration of the brain's local state-space geometry, distinct from and prior to any downstream consolidation outcome (Langell, 2025).

An internal single-dataset analysis of BOAS (ds005555) previously reported that spindle-rich N2 windows differed from matched non-spindle N2 controls on several stratified geometry endpoints, most consistently `d`, `d_s`, `d_n`, `e_m`, and `d_l` (plan documents, `articles/RVC_SLEEP/plan/plan1.md`). That result raised three open questions that motivate the present study: (1) does the effect replicate in an independent dataset with different hardware, staging protocol, and spindle-density distribution; (2) is the effect a genuine peri-event **timing** signature, or could it instead be driven by pre-existing differences between the spindle-adjacent and matched-control sleep baselines; and (3) does pre-event bodily (cardiac/respiratory) state - the Embodied Anchor Principle (EAP) side of NDT - moderate the size of the effect, which would tie a purely EEG-defined event to interoceptive/autonomic state.

This study answers all three questions using two rounds of analysis. First, ANPHY (HD-EEG) and BOAS (PSG) were run through an identical, pre-registered subject-weighted contrast protocol (Methods) that explicitly separates the **timing** question (event vs. pre-spindle baseline, within the same spindle) from the **specificity** question (event vs. matched non-spindle control), because the two ask different things and, as shown below, can even disagree in sign for the same endpoint if the pre-spindle baseline itself differs by dataset. Second, a third, independent PSG dataset (RichSleep) with simultaneous, near-complete cardiac **and** respiratory recording was added specifically to (a) replicate the timing effect out-of-sample and (b) run the cleanest EAP-moderation test attempted in this line of work: unlike the cardiac-only (ANPHY) and respiratory-only, majority-missing (BOAS) anchors used previously, RichSleep's anchor state combines both channels with near-100% subject coverage, removing the missingness objection that could previously explain a null moderation result.

Core question

Does the spindle-locked state-space reconfiguration reported in a single PSG dataset replicate across independent datasets when timing and specificity are analyzed separately, and is the effect moderated by pre-event embodied (cardiorespiratory) anchor state once that state can be measured without single-channel missingness?

2. Theory

2.1. Theoretical framework and primary metric

NDT models a latent noetic state X_t evolving under a geometry-shaped stochastic drift, $dX_t = -G_t(X_t)\nabla\Phi(X_t, t)dt + \sigma(t)dW_t$, and is measured through a versioned chart that projects raw electrophysiology into a low-dimensional Meta-Noetic Phase Space (MNPS). The canonical 3D chart $x_t = [m_t, d_t, e_t]^T$ separates **metastability/mobility** (m), **deviation from optimal integration-segregation balance** (d), and **entropy/entropic energy** (e). Each axis is additionally decomposed into a 9D stratified chart ,

$$x_t^{(9)} = [m_a, m_e, m_o, d_n, d_l, d_s, e_e, e_s, e_m]^T, \quad x_t = P_{\text{fixed}} x_t^{(9)} \tag{1}$$

where P_{fixed} is a release-fixed, auditable projection (not a claim of unique ontology). The nine stratified subcoordinates used as the primary endpoint family here are: m_a (attention-aligned precision pulses), m_e (rhythmic entrainment), m_o (spectral-centroid variability), d_n (heat-kernel-derived network-integration deviation), d_l (local clustering-based segregation load), d_s (representational-geometry deviation), e_e (sample entropy), e_s (permutation entropy), and e_m (Hjorth-mobility/beta-alpha composite). All nine, plus the three composite axes m, d, e, form the frozen 9-endpoint primary family used throughout (Methods, `S1.2`).

2.2. Embodied Anchoring Principle (EAP)

NDT's Embodied Anchoring Principle proposes that bodily/interoceptive rhythms (cardiac, respiratory, ocular) can condition the same latent landscape that governs cortical state-space dynamics, so that pre-event bodily state could in principle **moderate** how strongly a cortical event (here, a spindle) reconfigures MNPS geometry. Each NMD run co-exports an `anchor_state` layer alongside the EEG-only MNPS coordinates. Two flavors of anchor coverage are contrasted here: single-channel anchors (ANPHY: cardiac HRV only; BOAS: respiratory regularity only, majority-missing) and RichSleep's `cardioresp_anchor_index`, a composite built from **simultaneous**, near-fully-covered ECG and respiratory channels - closer to a genuine dual-arm embodied anchor than either single-channel predecessor. EAP is tested here strictly as a **moderator** of an EEG-only geometric effect, never as a component of the primary geometry itself (Methods, dual-surface contract).

3. Methods

3.1. Dataset

Three independent overnight PSG datasets were used, none with any experimental manipulation (purely observational):

- **ANPHY** ((Wei et al., 2024), (ANPHY Lab, 2024)) - 27 healthy adults, high-density scalp EEG (Biosemi, 93 channels), whole-night PSG.
- **BOAS / ds005555** ((López-Larraz et al., 2025)) - 128 subjects, PSG (Bitbrain Open Access Sleep dataset, OpenNeuro).
- **RichSleep** ((RichSleep contributors, 2026)) - 24 subjects, whole-night PSG with six EEG channels (F3-A2, F4-A1, C3-A2, C4-A1, O1-A2, O2-A1), two EOG channels (EOG-L, EOG-R), chin EMG, ECG, respiratory/snoring/leg-movement/blood-oxygen/carbon-dioxide channels, chest/abdominal movement, and breath-interval duration; 200 Hz sampling; impedance kept below 10 Ω during acquisition; hypnogram stages W/N1/N2/N3/R. RichSleep was added specifically as an out-of-sample verification dataset with a richer, near-complete simultaneous cardiac+respiratory anchor (Results, R6).

Sleep-spindle events were detected with YASA ((Vallat & Walker, 2021), C3-A2 channel, 200 Hz, sigma 12-15 Hz, N2 only) and exported as `sleep_spindle` events with per-event onset, duration, and peak frequency.

3.2. NMD pipeline

All three datasets were processed through the same NeuralManifoldDynamics (NMD) `summarize` pipeline, using a `primary_eeg_only` MNPS coordinate contract (`rvc_v1_eeg_only_e_m`: $e_m = 0.6 \cdot \text{eeg_hjorth_mobility} + 0.4 \cdot \text{eeg_beta_alpha}$, identical weights in all three datasets) so that the composite geometry itself never contains body-signal information. ANPHY and BOAS were additionally run through a second, **anchor-augmented** configuration in which e_m's EEG term is partially or fully replaced by the dataset's primary body anchor (ANPHY: 80% HRV RMSSD + 20% Hjorth mobility; BOAS: 70% respiratory regularity + 30% beta/alpha) - used only for the Pass B surface-sensitivity check (R5), never for the primary claim. RichSleep has no anchor-augmented counterpart; it is used only on its `primary_eeg_only` surface. Each subject's per-epoch export is aligned into 30 s windows and event-locked around each detected spindle into five bins (`pre_far`, `pre_near`, `event`, `post_near`, `post_far`) plus a `matched_control` condition (three non-spindle N2 controls per event, same subject, ≥ 30 s exclusion margin from any spindle/K-complex/arousal).

3.3. Primary endpoints

The frozen primary endpoint family is the nine stratified coordinates plus the three composite axes: $m, d, e, d_n, d_l, d_s, e_e, e_s, e_m$ (9 total; the three composites are algebraic recompositions of the six remaining stratified coordinates and are retained for interpretability, not as independent tests). Reachability-cone and MNJ endpoints were specified in the analysis plan (`plan/plan2.md`) as secondary layers to be added only if stable, but were not run under the current freeze and are not reported here.

3.4. Statistical design

The biological unit of analysis is the **subject**, not the event, because both datasets contribute highly uneven numbers of spindle events per subject (ANPHY median ≈ 40; BOAS median 5, range 1-18). For every (endpoint, dataset) cell: (1) an event-level delta is computed (either `event_bin - matched_control_median`, aggregated per event, or `event_bin - mean(pre_far bins, same event)`); (2) event-level deltas are collapsed to one **subject-mean delta**; (3) the primary p-value is a subject-level **sign-flip permutation test** ($n = 2000$ permutations) on the grand mean of subject-mean deltas - chosen because it directly tests the within-subject direction question without assuming a parametric distribution; (4) a one-sample t-test and Wilcoxon signed-rank on the same subject means are reported as sensitivity checks; (5) Benjamini-Hochberg FDR (Benjamini & Hochberg, 1995) is applied fresh within each frozen test family (9 endpoints $\times$ number of datasets in that family) at $\alpha = 0.05$.

3.5. Robustness protocol

1. **Subject-weighted vs. event-weighted estimand** - the primary estimand (A1r/A2r) subject-weights every event-count-imbalanced subject equally; an event-weighted cluster-robust OLS analysis (A1-original) is retained only as a pre-amendment sensitivity table.
2. **A1/A2 decomposition** - `event - control = (event - pre_far) + (pre_far - control)`, used to test whether an event-vs-control effect is a genuine timing effect, a pre-existing baseline-separation effect, or both (R4).

3. **Surface sensitivity (Pass B)** - comparing primary vs. anchor-augmented geometry on the identical subject-weighted estimand, to confirm the anchor-augmented surface behaves as a construction should (only the directly-modified coordinate moves) rather than as an implicit EAP test.
4. **EAP moderation (Pass C)** - `delta ~ anchor_pre_far + <mean-centered nuisance covariates>`, weighted inverse to each subject's event count, cluster-robust SE by subject, with a companion per-subject-slope sign-flip permutation test immune to event-count imbalance by construction.
5. **Cross-dataset replication (RichSleep)** - the identical A1r/A2r/Pass-C machinery re-run on an independent third dataset, with explicit sign-agreement reporting against the frozen ANPHY/BOAS tables (R6).

3.6. Secondary analysis

The RichSleep EAP-moderation test additionally reports four sensitivity anchors beyond the primary dual-arm composite (`anchor_index`, `anchor_index_v02`, `sympathetic_index`, `resp_anchor_index`, `ocular_stability_index`) to check whether any single composite index behaves differently from the mechanistically-motivated primary anchor; these are exploratory and are flagged where their coefficients are numerically negligible despite nominal FDR significance (R6).

4. Results

4.1. R1: Data quality and coverage

Dataset	N subj.	N spindle events	N control rows	Primary anchor	Anchor coverage
ANPHY	27	1,076	34,263	`ecg_hrv_rmssd_ms`	100%, 27/27
BOAS (ds005555)	128	634	192,312	`resp_regular_index`	34%, 44/128 (pre_far); Pass C $N \approx 35$
RichSleep	24	46,058 (raw, all bins); 3,900 after N2/coords filter	32,931	`cardioresp_anchor_index`	100%, 24/24

Table 1: Sample sizes, event/control coverage, and primary-anchor coverage per dataset. ANPHY/BOAS counts are raw event-locked rows for `bin_label="event"/"control"` (`event_locked_coverage.csv`); RichSleep counts are the `condition` totals from its event-locked parquet on the corrected ingest (§ Data, `data/raw/neuralmanifolddynamics_RichSleep_20260708_135627` - see R6′s data-provenance note for the earlier, buggy `..._063231` run this replaced). BOAS respiratory-anchor coverage reflects the 66/128 subjects with no respiratory channel documented in the handover. All 24 RichSleep subjects contribute to both A1r and A2r on the corrected ingest (Table Table 5).

Config-hash and subject-count sanity checks passed for all three datasets (ANPHY/BOAS: `run_manifest_check.csv`; RichSleep: `sanity_check.csv`, config digest `8cb255aa7055444b`, $N = 24$). RichSleep's `cardioresp_anchor_index`, `anchor_index`, `anchor_index_v02`, `sympathetic_index`, `vagal_index`, `resp_anchor_index`, and `ocular_stability_index` all show 0% missingness across all 24 subjects after the pass-A base filter - a substantially cleaner anchor-coverage profile than either single-channel predecessor.

4.2. R2: Event-vs-matched-control specificity (A1r)

Spindle-event windows differ from matched non-spindle N2 controls on 8 of 9 endpoints in the same direction in both ANPHY and BOAS, most surviving BH-FDR (Table Table 2). The exception is `d_n`, whose sign differs between datasets - resolved by decomposition in R4.

Endpoint	ANPHY Δ	q	BOAS Δ	q	Agreement
m	$+0.430$	< 0.001	$+0.230$	< 0.001	same sign
d	$+0.203$	< 0.001	$+0.404$	< 0.001	same sign
e	$+0.100$	< 0.001	$+0.274$	< 0.001	same sign
d_n	-0.189	< 0.001	$+0.589$	< 0.001	**opposite sign**
d_l	$+0.392$	< 0.001	$+0.349$	< 0.001	same sign
d_s	$+0.500$	< 0.001	$+0.427$	< 0.001	same sign
e_e	$+0.049$	0.082 (ns)	$+0.103$	0.0016	same sign
e_s	$+0.077$	0.0016	$+0.485$	< 0.001	same sign
e_m	$+0.488$	< 0.001	$+0.397$	< 0.001	same sign

Table 2: Pass A1r: subject-weighted event-vs-matched-control delta (subject-mean of `event_bin - matched_control_median`), sign-flip permutation p, BH-FDR q across the frozen 18-test family. Source: `results/rvc_sleep_20260703/02c_pass_a1r_subject_weighted_specificity/`.

Read together with R4, this broad specificity result is real but should not be read as a single clean "spindle-specific" mechanism - see the decomposition below.

4.3. R3: Within-spindle timing contrast (A2r) - primary claim

The within-spindle timing contrast (event bin vs. the same spindle's own `pre_far` baseline) is the cleanest and most consistent result: `d_n` is the only endpoint that is FDR-significant, same-sign, in both ANPHY and BOAS (Table Table 3), and (R6) replicates a third time in RichSleep.

Endpoint	ANPHY Δ	q	BOAS Δ	q
d_n	$+0.106$	0.009	$+0.165$	0.011
e_s	$+0.062$	0.062 (ns)	$+0.114$	0.011
d	$+0.040$	0.037	$+0.072$	0.014
e_e	-0.064	0.009	-0.019	0.57 (ns)
all others	ns	-	ns	-

Table 3: Pass A2r: subject-balanced event-vs-pre_far within-spindle timing delta. Source: `results/rvc_sleep_20260703/03b_pass_a2r_subject_balanced_profile/`.

Primary finding

Sleep spindles are followed by a reproducible rise in d_n relative to their own pre-spindle baseline, in every dataset tested (ANPHY, BOAS, and - R6 - RichSleep). This within-spindle timing signature, not the broader event-vs-control contrast, is treated as the study's primary claim.

4.4. R4: Decomposition - resolving the `d_n` sign conflict

Decomposing the event-vs-control delta as $C(\text{event} - \text{control}) = A(\text{event} - \text{pre_far}) + B(\text{pre_far} - \text{control})$ shows that the two datasets **agree** on the timing component (A, both positive) but **disagree** on how each dataset's own pre-spindle baseline compares to its matched generic-N2 controls (B: ANPHY -0.293, BOAS $+0.427$) - which fully explains why the A1r-original `d_n` sign flips between datasets while the A2r timing sign does not (Table Table 4). `d_s` and, to a lesser extent, `m` show the opposite pattern: their event-vs-control effect is driven almost entirely by B (baseline separation), with a near-zero or slightly negative timing component A.

Endpoint	ANPHY A / B / C	BOAS A / B / C	Reading
d_n	$+0.106/-0.293/-0.187$	$+0.166/+0.427/+0.593$	A agrees (timing rise, both $+$); B disagrees by dataset - explains the A1r sign conflict
d_s	$+0.018/+0.484/+0.501$	$+0.066/+0.354/+0.420$	mostly B (baseline separation), weak/no timing component
e_s	$+0.062/+0.017/+0.079$	$+0.116/+0.368/+0.484$	A positive both datasets; B small in ANPHY, large in BOAS
m	$-0.013/+0.442/+0.429$	$-0.041/+0.295/+0.254$	timing component near-zero/negative both datasets; result is baseline-driven

Table 4: Pass A1 decomposition: A = timing component (event−pre_far), B = pre-spindle-baseline vs. matched-control separation, $C = A + B$ = the A1r event-vs-control delta. Source: `results/rvc_sleep_20260703/02d_pass_a1_decomposition/`.

4.5. R5: Embodied-anchor moderation is not supported (Pass B / Pass C)

Pass B (surface-sensitivity, construction-validity check, not an EAP test). Comparing primary vs. anchor-augmented geometry on the identical subject-weighted estimand: 7 of 9 endpoints (`m, d, d_n, d_l, d_s, e_e, e_s`) are byte-identical between surfaces (surface delta $= 0$, $r = 1.0$) in both datasets and both delta modes - expected, since these coordinates never touch body-signal input. Only `e_m` (directly modified by anchor injection) and, more weakly, `e` (which recomposes from `e_m`) show any divergence (e_m: $r = 0.36$-0.74; e: $r \geq 0.97$ everywhere). FDR-significance status changes in **0 of 36** testable rows. Per the pre-registered guardrail, this shows the anchor-augmented surface is sensitive to injected body signals exactly where expected and provides no information about whether pre-event anchor state moderates EEG-only spindle geometry.

Pass C (the moderation test itself). Modeling `delta ~ anchor_pre_far + nuisance covariates`, weighted inverse to subject event count, cluster-robust SE, across 2 delta modes × 3 anchor models × 9 endpoints (54 tests): **53/54 do not survive FDR.** The single nominal exception (ANPHY `d_l`, HRV anchor, event-control delta, pooled $q = 0.0015$) does not replicate in its own subject-level-slope sensitivity check ($p = 0.140$, ns) and is treated as a pooled-model weighting artifact rather than a genuine effect.

Dissociation finding

Anchor-augmented geometry moves only where it structurally must (e_m, weakly e) and never flips an FDR conclusion (Pass B); the direct moderation test (Pass C) is null on 53/54 comparisons in two independent single-channel anchors. Spindle-locked geometry reconfiguration and pre-event cardiac/respiratory anchor state are dissociated in this dataset pair.

4.6. R6: RichSleep - three-dataset replication and a “proper” dual-arm EAP test

Data provenance note

An earlier ingest of RichSleep (`neuralmanifolddynamics_RichSleep_20260704_063231`) had a subject-scoping bug in event-locked CSV-source resolution that silently assigned one subject’s spindle catalog to all 24 “subjects” (detected via a signature check: all 24 shared an identical first-three-onset timing pattern, vs. 27/27 and 122/125 distinct signatures in ANPHY and BOAS respectively, which are unaffected). This was traced, fixed, and re-run (`neuralmanifolddynamics_RichSleep_20260708_135627`, confirmed 24/24 distinct per-subject spindle catalogs, count range 161-2,562, consistent with genuine per-subject YASA detection). All R6 numbers below are from the corrected run; see diary 233-234 for the full trace.

RichSleep was added specifically to (a) test whether R3′s primary timing claim survives in a fully independent PSG dataset, and (b) run the EAP-moderation test with an anchor that does not carry ANPHY’s cardiac-only or BOAS’s majority-missing respiratory limitation. RichSleep’s `cardioresp_anchor_index` is built from simultaneous ECG and respiratory channels with 0% missingness across all 24 subjects (Table Table 1), so a null result here cannot be attributed to single-channel missingness.

A1r replication (event vs. matched-control, $N = 24$ subjects, 3,615 events). 7 of 9 endpoints sign-agree with the frozen ANPHY/BOAS direction, 8 of 9 FDR-significant (Table Table 5); the two headline endpoints from R3, `d_n` ($+0.29$, $q = 0.0006$) and `e_s` ($+0.22$, $q = 0.0006$), replicate with large effect sizes. `m` now agrees in sign with ANPHY/BOAS (resolving the discrepancy noted on the earlier, buggy ingest); `e` loses FDR significance on the corrected data ($q = 0.14$); `e_e` disagrees in sign (FDR-significant, opposite direction) - a genuine open discrepancy, not smoothed over here.

Endpoint	RichSleep Δ	q	Sign agrees w/ ANPHY+BOAS
`m`	$+0.230$	0.0006	Yes
`d`	$+0.349$	0.0006	Yes
`e`	$+0.019$	0.139	Yes (ns)
`d_n`	$+0.292$	0.0006	**Yes**
`d_l`	$+0.433$	0.0006	Yes
`d_s`	$+0.148$	0.0006	Yes
`e_e`	-0.222	0.0006	No
`e_s`	$+0.223$	0.0006	**Yes**
`e_m`	$+0.556$	0.0006	Yes

Table 5: RichSleep A1r replication vs. the frozen ANPHY/BOAS A1r direction, on the corrected ingest. Source: `results/rvc_sleep_20260708/09_richsleep_verification/richsleep_a1r_specificity.csv`.

A2r replication (event vs. pre_far, $N = 24$ subjects, 160 paired events). The primary timing endpoints still do not survive FDR on the corrected, larger sample (`d_n`: $q = 0.72$; `e_s`: $q = 0.71$), and only 5/9 endpoints sign-agree (`m` and `e` now disagree). Two non-primary endpoints newly reach FDR significance with more events than the earlier ingest had (`e_e`: $q = 0.0045$; `e_m`: $q = 0.036$). Given the still-modest paired-event count relative to ANPHY/BOAS (160 vs. 1,037/568) and that the two pre-registered headline endpoints remain null, this is read as **underpowered and inconclusive for the primary timing claim**, not as a disconfirmation of R3 - and the two endpoints that do reach significance here are not the ones the manuscript leads with.

EAP moderation, primary dual-arm anchor (`cardioresp_anchor_index_pre_far`, $N = 24$ subjects, 2,260-2,435 events, both delta modes). A clean null across all 9 endpoints in both delta modes (18 tests; p range ≈ 0.04-0.94, no FDR survivors, min $q \approx 0.32$) - the same qualitative result as Pass C's ANPHY/BOAS null, now obtained with the full, correctly-scoped 24-subject sample on the one dataset where the missingness objection does not apply. Several single-channel sensitivity anchors (`anchor_index`, `anchor_index_v02`, `sympathetic_index`, `ocular_stability_index`) show nominally FDR-significant but numerically negligible coefficients ($\approx 10^{-8}$-10^{-9} per raw unit of a large-scale composite index); this is read as a scale/multicollinearity artifact of those raw composites rather than genuine moderation, the same caution already applied to BOAS's `anchor_index_v02` sensitivity model. One sensitivity anchor does **not** fit this pattern: `resp_anchor_index` shows FDR-surviving coefficients of ordinary, non-negligible magnitude (≈ -0.19 to -0.36) on `m`, `d`, `d_l`, and `e_m`, corroborated by the companion subject-level-slope permutation test for three of the four ($p \approx 0.0005$-0.004). This is **not** explained by the scale-artifact reasoning used for the other sensitivity anchors and is reported here as an open, unresolved follow-up question rather than dismissed (S2.5) - it does not change the primary anchor's null result, but should not be over-read as fully consistent with it either.

Three-dataset synthesis

Three independent PSG datasets, spanning cardiac-only, respiratory-only, and dual-arm cardiorespiratory anchor measurement, now agree: spindle-locked d_n (and, more weakly, e_s) reconfiguration replicates, and no tested embodied-anchor signal moderates it.

4.7. R6b: RichSleep phase-resolved EAP mechanism test

R6′s `cardioresp_anchor_index_pre_far` moderation test uses a tonic (multi-minute pre-event average) anchor as a proxy for embodied-anchor state - a coarse test of the mechanism EAP actually proposes (phase-resolved cardiorespiratory entrainment of the denoising process, `@eq:rvc` in the NDT framework). Using NMD's phase-resolved RichSleep export (`event_phase_v2`, independently verified 24/24-subject onset-catalog match against the corrected event-locked run before use), we ran two more direct, mechanistically-motivated tests on the same

N2-staged, geometry-valid spindle population R6 uses (190 events, 24 subjects; this population is considerably smaller than R6′s full-bin event counts because the `event`-bin window's stage label does not always coincide with N2 despite N2-only spindle detection - an open, unresolved discrepancy, not investigated further here).

Mål B (self-prior/boundary-condition test): does spindle onset preferentially occur at a particular cardiac or respiratory phase? A per-subject Rayleigh test found no within-subject phase-locking for either channel (0/24 subjects FDR-significant for cardiac onset phase, 0/24 for respiratory onset phase). A second-order Rayleigh test on the 24 subject-level mean directions found no population-level common preferred phase either (cardiac: $R = 0.160$, $p = 0.547$; respiratory: $R = 0.109$, $p = 0.755$).

Mål C (HEP-amplitude moderation): does pre-event heartbeat-evoked-potential amplitude moderate the spindle-locked geometry response? Using `hep_amplitude_pre_far` - a more direct interoceptive-precision proxy than the tonic composite anchor - as the sole predictor in the same WLS-cluster + subject-level-slope-permutation machinery as R6/Pass C, across both delta modes and all 9 endpoints (18 tests): a clean null (p range $\approx$ 0.10-0.96, 0/18 FDR survivors, corroborated by the companion slope-permutation test).

R6b summary

The fairest test of the EAP mechanism attempted in this study - direct phase-locking and HEP-amplitude moderation, rather than a tonic-state proxy - is also null. This strengthens rather than weakens R6′s conclusion: it is not merely that a coarse anchor proxy failed to detect moderation, but that two more mechanistically direct routes to the same claim find nothing either, on the one dataset with the richest simultaneous cardiorespiratory instrumentation. Given the modest event count (190 events, 24 subjects), this should be read as **no effect detected at this power**, not as definitive disconfirmation of the EAP mechanism in general (S2.6, S3.4e; diary 236).

5. Discussion

5.1. Primary finding

The within-spindle timing rise in d_n - a state-space endpoint most directly tied to network integration-segregation balance - is the single most consistent result in this study: it replicates in direction and FDR significance across three independent PSG datasets collected with different hardware, staging protocols, and spindle-density distributions (ANPHY, BOAS, RichSleep). This is a stronger evidentiary position than the original single-dataset (BOAS) observation that motivated the study, and supports treating spindle events as genuine, geometrically detectable Rhythmic Variance Control events rather than a dataset-specific artifact of one detector or one recording montage.

5.2. Confound interpretation

The broader event-vs-matched-control contrast (A1r, R2) is real on most endpoints but is not, by itself, a clean single mechanism: the decomposition (R4) shows that for `d_n` specifically, the apparent cross-dataset disagreement in A1r sign is fully explained by a dataset-specific difference in how each cohort's **pre-spindle** baseline relates to its matched generic-N2 controls, not by any disagreement in the **timing** signature. For `d_s` and `m`, the event-vs-control effect is mostly a baseline-separation effect with little or no timing component. This is why the manuscript leads with the timing contrast (A2r/R3/R6) rather than the broader specificity contrast (A1r/R2) as the primary claim - the latter should be read as "broad, present, and baseline-sensitive," not as direct evidence of a single spindle-specific geometric signature across every endpoint.

5.3. Secondary findings

Pass B's surface-sensitivity result (R5) is a construction-validity check, not an EAP finding: it confirms the anchor-augmented geometry export behaves exactly as intended (only the directly-modified coordinate, and weakly its recomposition, respond to injected body signal) and never changes an inferential conclusion. Pass C and its RichSleep replication are read together as a genuine negative result for embodied-anchor moderation of spindle-locked geometry, under the anchors and models tested here - not a failure of the analysis, and not evidence that EAP-style body-brain coupling is absent in general, only that the specific moderation hypothesis tested (pre-event cardiorespiratory anchor state predicting spindle-locked geometry change) was not supported across three increasingly well-instrumented tests of it.

5.4. Limitations

- BOAS's respiratory-anchor Pass C model runs on a restricted N ($\approx$ 35/128 subjects with a valid pre-event `resp_regular_index`); the larger-N sensitivity anchor (`anchor_index_v02`, $N \approx 90$) is still null, but this

does not fully remove the missingness concern for BOAS specifically - RichSleep's dual-arm anchor was added in part to address this.

- RichSleep's A2r within-spindle timing replication remains underpowered relative to ANPHY/BOAS ($N = 24$ subjects, 160 paired events vs. 1,037/568) and the two pre-registered headline endpoints (`d_n`, `e_s`) are not FDR-significant here; this is reported as inconclusive rather than as a second disconfirmation of R3.
- The `e_e` sign discrepancy between RichSleep and ANPHY/BOAS in the A1r contrast (R6) is a genuine open finding that has not yet been investigated at the subject level; RichSleep's more fragmented sleep-stage architecture (documented in its ingest configuration) is one candidate explanation, not yet tested. (The earlier draft additionally flagged an `m` sign discrepancy; that was an artifact of the RichSleep ingest bug described in R6′s data-provenance note and is resolved - `m` now agrees in direction - on the corrected data.)
- RichSleep's `resp_anchor_index` sensitivity model (R6) shows FDR-surviving, non-negligible coefficients on several endpoints, partly corroborated by the subject-level-slope permutation check - unlike the other sensitivity anchors, this is not readily dismissed as a scale artifact. It does not overturn the primary `cardioresp_anchor_index` null but is an open follow-up (S2.5), not yet investigated further (e.g. whether it reflects genuine single-channel respiratory moderation masked in the composite anchor, or a confound with the primary anchor).
- Reachability-cone and Meta-Noetic Jacobian endpoints were specified as planned secondary layers but were not run under the current analysis freeze.
- `frequency_hz` and sigma-power residualization of the spindle detector are not available in the current NMD event-locked export schema and are not silently substituted for; this remains an open follow-up for detector-strength robustness.
- All three datasets are healthy-adult, observational, cross-sectional samples; no clinical population, intervention, or longitudinal design is tested (see `S4_Supplementary_Clinical.typ`).

5.5. Conclusion

Across three independent PSG datasets and 179 subjects, spindle events are reliably followed by a rise in the d_n state-space endpoint relative to their own pre-spindle baseline - the most defensible, dataset-independent claim this study supports. The broader event-vs-matched-control specificity result is real but baseline-sensitive and should not be over-read as a single unified mechanism. Pre-event embodied (cardiac and/or respiratory) anchor state does not moderate this effect under any anchor tested, including a genuinely dual-arm, near-fully-covered cardiorespiratory composite that removes the single-channel-missingness objection available to the two earlier null results.

6. Acknowledgements

The author thanks the ANPHY-Sleep, BOAS (ds005555), and RichSleep dataset contributors for open data access, and the YASA project for the spindle-detection tooling used throughout.

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Author Contributions

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Funding Information

Funding information will be inserted at submission.

Data Availability Statement

ANPHY-Sleep is openly available via OSF ((ANPHY Lab, 2024), described in (Wei et al., 2024)). BOAS (ds005555) is openly available via OpenNeuro ((López-Larraz et al., 2025)). RichSleep is described in (RichSleep contributors, 2026); formal access details will be added at submission. NMD (NeuralManifoldDynamics) pipeline outputs and all analysis scripts are frozen and versioned in the analysis repository under `articles/RVC_SLEEP/src/`, with per-pass result tables under `articles/RVC_SLEEP/results/`.

Competing Interests

The author declares no competing interests, subject to final confirmation at submission.